

SPOTTER

Freight-Rate Machine Learning Assessment

Final model selection, validation evidence, predictions, and reproducibility

Main submission

Full-feature direct-target Ridge ($\alpha=10$) trained on all 48,000 labeled rows.

December chart

Separately validated December-compatible CatBoost: 100 trees, depth 8, learning rate 0.10, seed 42, CPU.

Evidence-led selection | Chronological validation | Immutable source inputs | Scorer-ready outputs

01 Executive decision

The main submission uses Ridge because it led the declared four-fold temporal ranking: mean MAE \$152.17, worst-fold MAE \$174.05, mean RMSE \$633.64, and MAE standard deviation \$14.77. Advanced CatBoost candidates improved October and tail slices but were less stable across months.

The December output uses a separate compatible CatBoost model. It reduced rolling MAE from \$203.59 for compatible Ridge to \$167.72 while avoiding fabricated market, quote, coordinate, or route inputs.

Outcome

Artifact	Contract	Status
Main predictions	12,000 template-aligned IDs	PASS
December predictions	31 dates; original 7 columns	PASS
Repository tests	19 contract and robustness tests	PASS
Employer scorer	Both CSV contracts and chart	PASS

Final metrics are calculated by Spotter after submission; no hidden validation target was available or used.

Why two models?

Assessment validation supplies market and quote fields, while the fixed December file does not. Separate role-specific models preserve the strongest evidence available for each exact prediction contract.

02 Data, quality, and leakage controls

The labeled data contains 48,000 loads from January through October 2025. Assessment validation contains 12,000 November-December loads. The chart input contains one fixed Lexington-to-Fort Wayne Dry Van load for every December date.

Dataset	Rows	Period	Role
Labeled training	48,000	2025-01-01 to 2025-10-31	Model development and final fit
Assessment validation	12,000	2025-11 to 2025-12	Main predictions; target hidden
December chart inputs	31	2025-12-01 to 2025-12-31	Fixed-lane daily predictions

Quality audit

Schemas, row counts, identifier uniqueness, date ranges, missingness, duplicate rows, numeric coercion, infinite values, categorical cardinality, outliers, and train-to-validation shifts were checked. The seven employer files remain byte-identical to the recorded SHA-256 manifest.

Leakage controls

load_id and posted_rate are prohibited model inputs. All preprocessing is fitted inside each temporal training fold. Date and domain features are deterministic and target-free. No validation target is fitted, inspected, or imputed.

Ridge applies fold-fit numeric medians, missing indicators, scaling, constant categorical imputation, and unknown-safe one-hot encoding. CatBoost handles native categoricals and missing values without dropping rows.

03 Chronological validation

Random splitting would leak future market regimes into training. Four expanding windows recreate the forward-looking prediction setting and allow stability to influence selection.

Fold	Training	Validation	Train rows	Validation rows
July	Jan-Jun	July	28,806	4,912
August	Jan-Jul	August	33,718	4,759
September	Jan-Aug	September	38,477	4,670
October	Jan-Sep	October	43,147	4,853

Ranking criteria

Selection prioritized rolling mean MAE, then worst-fold MAE and fold stability. October performance, RMSE, WMAPE, tail behavior, unseen categories, and operational feature availability were secondary but explicit checks.

Metrics

Metric	Interpretation
MAE	Typical absolute dollar error; primary ranking
RMSE	Emphasizes rare large misses
WMAPE	Portfolio absolute error relative to portfolio value

The target is right-skewed. A low MAE alone can obscure severe underprediction of rare expensive loads, so tail and business slices are mandatory.

04 Model comparison and selection

Model / policy	Mean MAE	SD	Worst	Oct MAE	RMSE	WMAPE
Ridge full	152.17	14.77	174.05	143.47	633.64	6.379%
CatBoost market-only	153.47	32.21	201.27	141.45	638.03	6.429%
CatBoost full	158.44	39.39	215.15	134.52	640.78	6.639%
CatBoost no-signal	161.16	36.61	213.63	149.13	638.30	6.753%
CatBoost Dec-compatible	167.72	42.96	228.44	144.68	643.34	7.029%
Ridge Dec-compatible	203.59	37.13	251.44	170.63	642.63	8.549%

Main-model decision

Market-only CatBoost nearly tied mean MAE and slightly improved October, but its worst fold was \$27.22 worse and its variability was more than twice Ridge's. Full CatBoost delivered the best October MAE but was weaker across the complete time series. Ridge therefore remains the defensible main model.

December decision

Compatible CatBoost improved mean MAE by \$35.87 (17.6%) and October MAE by \$25.95 (15.2%) over compatible Ridge. It uses only fields available in the fixed December contract.

Signal policy

market_index is useful within labeled temporal folds but its provenance and future availability are unconfirmed. It remains in the main model because assessment validation supplies it. It is excluded from December because fabricating future values would introduce an unsupported dependency.

05 Error analysis and robustness

October error slices show the key remaining business risks.

Slice	Rows	Ridge MAE	Interpretation
All October	4,853	\$143.47	Strong typical-load accuracy
Top 10% rate	505	\$595.98	Expensive loads remain compressed
Distance >2,000	769	\$284.09	Higher long-haul dollar risk
Reefer	1,190	\$173.13	Higher error than other equipment
Dry Van	2,745	\$133.24	Best-supported equipment segment

Advanced-model evidence

CatBoost improved expensive, long-distance, Reefer, and synthetic unseen-city slices, but it did not erase severe tail risk. Its top-10% RMSE remained about \$1,900 in October.

Unseen-city stress

A deterministic feature-only stress test removed five frequent endpoint cities entirely from training. CatBoost market-only achieved MAE \$121.85, compared with \$138.52 for Ridge. All pipelines accepted unknown categories without failure. This is robustness evidence, not an official temporal selection fold.

The final set contains eight new cities and 1,461 unseen routes - more exposure than the labeled time folds - so hidden-period error may differ from the backtest.

06 Final fit and prediction sanity

Both models were trained on all 48,000 labeled rows after selection was locked. Model files, parameters, package versions, output hashes, and positive-floor diagnostics are recorded in `final_model_metadata.json`.

Prediction set	Rows	Minimum	Mean	Median	Maximum
Assessment validation	12,000	\$33.61	\$2,422.00	\$2,085.55	\$6,590.50
Training target	48,000	\$57.22	\$2,373.98	\$2,030.76	\$25,533.00
December fixed lane	31	\$825.57	\$860.72	\$851.14	\$987.35

Sanity findings

Validation predictions rise coherently across distance bands and are highest on average for Reefer. November and December assessment means are close (\$2,418.76 and \$2,425.06). No raw final predictions were nonpositive, so the \$0.000001 safety floor changed zero values.

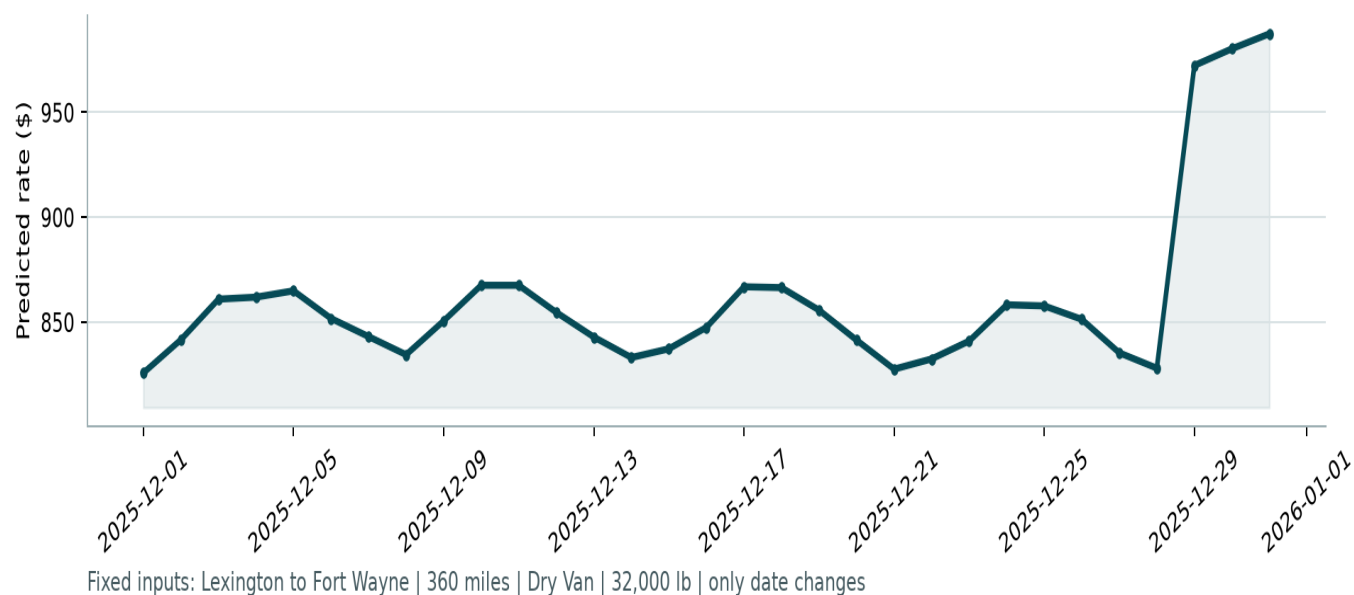
December behavior

The first five fixed-lane values are \$825.57, \$841.42, \$860.76, \$861.71, and \$864.73. The last three rise to \$972.20, \$980.19, and \$987.35. That year-end step is a calendar extrapolation from the trained model, not causal proof of a market event.

07 December fixed-lane chart

Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb | only date changes

Candidate: December 2025 Predicted Load Rate



The chart was generated by score.py after it validated the 12,000-row submission and the 31-row December contract. Rates mostly vary in a weekly pattern before the learned year-end step.

Interpretation guardrail

The December model is intentionally signal-free and compatible with the supplied fields. The curve is a model output under fixed load attributes; it is not a confidence interval, external market forecast, or guarantee of transactable rates.

08 Reproducibility, limitations, and conclusion

Reproducibility

The finalization script locates the repository, validates schemas, fits both locked models, aligns IDs one-to-one, writes new outputs, serializes models, and records SHA-256 hashes. Tests rerun inference from the saved artifacts and compare results at tight numerical tolerance.

Control	Result
Employer inputs	Unchanged; SHA-256 manifest matches
Feature exclusions	load_id and posted_rate absent
Main alignment	One-to-one merge in template order
Saved inference	Deterministic rerun passes
Scorer	Both files validated; chart created

Limitations

The official hidden metric is unknown. Rare expensive loads remain underpredicted. The final city/route mix is not fully represented by labeled folds. Main-model market signals have operational uncertainty. CatBoost tuning was purposefully CPU-bounded, although a focused larger refinement was worse.

Conclusion

The locked solution follows the strongest available evidence: stable full-feature Ridge for assessment validation and separately validated compatible CatBoost for the December chart. Both artifacts are positive, finite, schema-correct, scorer-valid, and reproducible.